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CHAPTER 1 • NUMBER • TOPIC 15

Direct & Inverse Proportion

Write the α , find k , write the link, then substitute. Every proportion question is the same four moves: L-I-N-K.



$$\frac{3}{10} = 0.3 =$$

30%**6**

VARIANTS

17

PAST-PAPER PARTS

47

PRACTICE QUESTIONS

YOU WILL MASTER

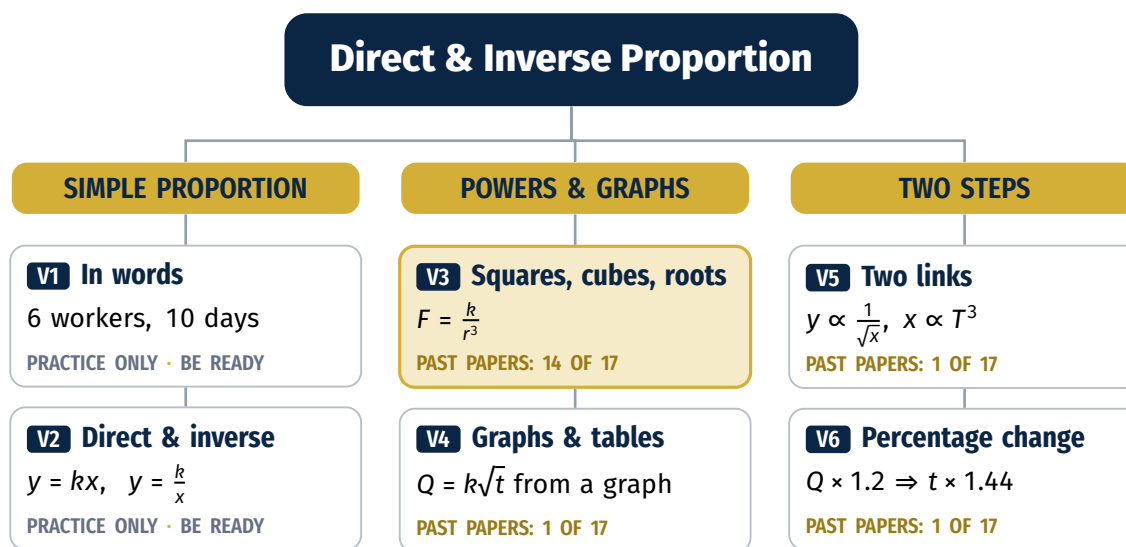
V1 In words V2 Direct & inverse V3 Squares, cubes, roots

V4 Graphs & tables V5 Two links V6 Percentage change

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Proportion at a glance



Your notes


★ How often does each variant come up?

13 past-paper questions (Oct 2020 → Nov 2025) hold 17 parts; 47 practice parts — grouped by type.

	PAST PAPERS (OUT OF 17)	PRACTICE PARTS
V1 In words	0	10
V2 Direct and inverse	0	7
V3 Squares, cubes, roots	14	11
V4 Graphs and tables	1	10
V5 Two links	1	4
V6 Percentage change	1	5

WHAT THIS TELLS YOU It is on almost every paper, near Q15–18, for 3–5 marks — and 14 of 17 parts are “proportional to the square / cube / root”. LINK it every time and the marks are yours.

BEFORE YOU START

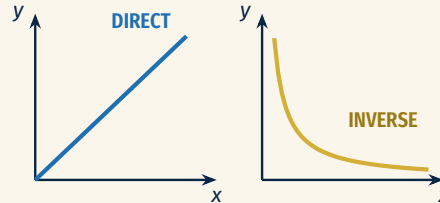
The big idea

Proportional means “one is a fixed multiple of the other”.

Direct: $y = kx$ — double x , double y .

Inverse: $y = \frac{k}{x}$ — double x , halve y .

The words “square”, “cube”, “square root” just change what x becomes: x^2 , x^3 , $\sqrt{x}$.



Your notes



The L-I-N-K method

L — Layout

$$F = \frac{k}{r^2}$$

Words →
equation with k .

I — Insert

$$36 = \frac{k}{16}$$

First pair in; find
 k .

N — New

$$F = \frac{576}{r^2}$$

The formula with
 k . Never round k .

K — Keep solving

$$F = \frac{576}{48^2}$$

Substitute the
new value.



Words → Layout

“directly ... square of d ”

$$B = kd^2$$

“varies as” means the
same.

“inversely ... cube of r ”

$$F = \frac{k}{r^3}$$

Inverse: k on top.

“inversely ... $\sqrt{m}$ ”

$$T = \frac{k}{\sqrt{m}}$$

Root goes on the bottom.

**CALCULATOR FIRST, THEN CHECK**

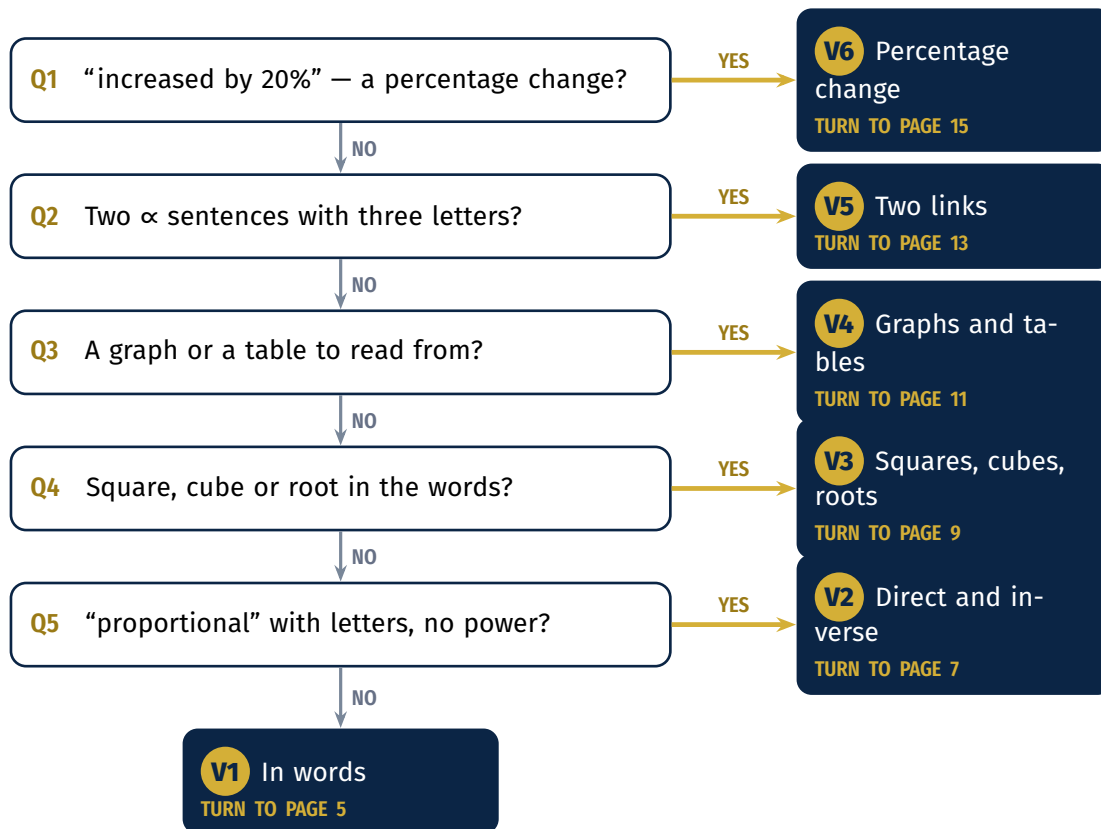
Check the formula with the first pair: $F = \frac{576}{4^2} = 36$ ✓. Keep k exact — a fraction is fine.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|--|-----------------|--|-----------------------|
| | What is it? | | When you see → you do |
| | How do I do it? | | Watch out |
| | Worked example | | Exam tip |
| | Check yourself | | Calculator check |

V1

VARIANT 1

Proportion in Words

More workers, less time: multiply to get the total job, then divide.

SEEN IN PAST PAPERS

0 / 17

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 10

 Your notes

? What is it?

In plain words

Direct (more pens, more cost): find the cost of **one**, then multiply.

Inverse (more workers, fewer days): find the **total job** — workers $\times$ days — then divide by the new number of workers.

IT LOOKS LIKE

5 pens £3.40

6 workers in 10 days

6 workers $\times$ 10 days

= 60 worker-days

4 workers: $60 \div 4 = 15$ days

How do I do it?

1 Direct or inverse?

More of one $\rightarrow$ more of the other? Or less?

2 Find one / total

Direct: $\div$ for one. Inverse: $\times$ for the total job.

3 Scale

Direct: $\times$ new amount. Inverse: $\div$ new amount.

4 Sense

More workers must mean fewer days.

INVERSE: TOTAL JOB FIRST workers $\times$ days, then share it out.

V1 Proportion in Words

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO*"5 pens cost £3.40. Cost of 8?"*→ **Direct: one pen, then $\times 8$** *"6 workers take 10 days. 4 workers?"*→ **Inverse: 6×10 , then $\div 4$** *"Working at the same rate" /
"assumption"*→ **Say: everyone works at the same
speed****WORKED EXAMPLE DR ESAM**

Proportion in words • 2 marks

6 workers build a wall in 10 days. How many days would 4 workers take, working at the same rate?

STEP 1 fewer workers $\Rightarrow$ more days ◀ inverse

STEP 2 $6 \times 10 = 60$ worker-days ◀ total job

STEP 3 $60 \div 4 = 15$ ◀ share it out

ANSWER 15 days

✓ more than 10 ✓

WATCH OUT

$$\times 10 \times \frac{4}{6} = 6.7 \text{ days}$$

$$\checkmark 10 \times \frac{6}{4} = 15 \text{ days}$$

Fewer workers can't finish sooner.

EXAM TIP

"Assumption" questions: every worker (or tap, or machine) works at the same rate.

Check yourself

calculator first • answers on page 17

1.1 Dr Eslam

5 pens cost £3.40. Work out the cost of 8 pens.

Answer:

1.2 Dr Eslam

3 taps fill a tank in 40 minutes. How long would 5 taps take?

Answer:

1.3 Dr Eslam

A car uses 6 litres for 75 km. How far on 20 litres?

Answer:

V2

VARIANT 2

Direct & Inverse: LINK

Layout, Insert, New equation, Keep solving.

SEEN IN PAST PAPERS

0 / 17

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 7
Your notes**? What is it?****In plain words**

“y is proportional to x” $\Rightarrow y = kx$. “y is inversely proportional to x” $\Rightarrow y = \frac{k}{x}$.

Use the **first pair** to find k , write the **formula**, then use it for the new value.

IT LOOKS LIKE

$$p \propto \frac{1}{q}$$

$$y \propto x$$

L

$$p = \frac{k}{q}$$

I

$$6 = \frac{k}{4}, k = 24$$

N

$$p = \frac{24}{q}$$

K

$$q = 3 : p = 8$$

**How do I do it?****1 L – Layout** $\propto \rightarrow$ equation with k .**2 I – Insert**First pair: find k .**3 N – New equation**Write the formula with k .**4 K – Keep solving**

Put in the new value.

L-I-N-K

“Find a formula” stops at N.

V2 Direct & Inverse: LINK

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“ y is directly proportional to x ” $\longrightarrow y = kx$

“ p is inversely proportional to q ” $\longrightarrow p = \frac{k}{q}$

“Find a formula for ...” $\longrightarrow$ **Stop after N – write the equation**

WORKED EXAMPLE DR ESLAM

LINK • 3 marks

p is inversely proportional to q . $p = 6$ when $q = 4$. Find p when $q = 3$.

STEP 1 $p = \frac{k}{q}$

◀ L: Layout

STEP 2 $6 = \frac{k}{4} \Rightarrow k = 24$

◀ I: Insert

STEP 3 $p = \frac{24}{q}$

◀ N: New equation

STEP 4 $p = \frac{24}{3} = 8$

◀ K: Keep solving

ANSWER $p = 8$

✓ q down, p up ✓**WATCH OUT**

✗ $p = kq$ for “inversely”

✓ $p = \frac{k}{q}$

Inverse puts the letter on the bottom.

EXAM TIP

Write the formula line ($p = \frac{24}{q}$) – it is a mark on its own.

Check yourself

calculator first • answers on page 17

2.1

Dr Eslam

$y \propto x$. $y = 20$ when
 $x = 4$. Find y when
 $x = 7$.

Answer:

2.2

Dr Eslam

w is directly
proportional to t ;
 $w = 12$ when $t = 3$. Find
a formula.

Answer:

2.3

Dr Eslam

d is inversely
proportional to c ;
 $d = 35$ when $c = 200$.
Find d when $c = 350$.

Answer:

V3**VARIANT 3** MOST COMMON IN THE EXAM

Squares, Cubes & Roots

Same LINK — the power or root goes on the letter in the Layout.

SEEN IN PAST PAPERS

14 / 17

PRACTICE QUESTIONS: 11



Your notes

Came up in: Oct/Nov 2020 P1H Q14 • May 2021 P2H Q16 • Jan 2022 P1H Q15 • Jun 2022 P2H Q17 • May 2023 P1HR Q17 • Jun 2024 P2H Q17 • Nov 2024 P1H Q14 • Jun 2025 P1HR Q18 • Jun 2025 P1H Q18 • Nov 2025 P2H Q17

? What is it?

In plain words

“Inversely proportional to the **cube** of r ” $\Rightarrow F = \frac{k}{r^3}$. “Directly ...

square root of d ” $\Rightarrow Q = k\sqrt{d}$.

Then LINK as before. Going backwards (find r) you

undo the power: cube root, square root.

IT LOOKS LIKE

$$F = \frac{k}{r^3}$$

$$M = kh^3$$

$$T = \frac{k}{\sqrt{m}}$$

$$F = \frac{k}{r^3}$$

$$6 = \frac{k}{8} \Rightarrow k = 48$$

$$3072 = \frac{48}{r^3} \Rightarrow r^3 = \frac{1}{64}$$

$$r = \sqrt[3]{\frac{1}{64}} = 0.25$$

How do I do it?

1 L — Layout

Power/root on the letter: kh^3 , $\frac{k}{r^2}$, $k\sqrt{d}$.

2 I — Insert

First pair; find k exactly.

3 N — New equation

Formula with k .

4 K — Keep solving

Substitute; undo the power if you are finding the letter.

POWER ON THE LETTER 14 of the 17 past-paper parts.

V3 Squares, Cubes & Roots

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"inversely proportional to the square of r " → $F = \frac{k}{r^2}$

" M varies directly as the cube of h " → $M = kh^3$

"Find the value of r when ..." → **Rearrange, then cube root / square root**

WORKED EXAMPLE **PAST PAPER**

June 2025 • Paper 1HR • Q18 • 5 marks

F is inversely proportional to the cube of r . $F = 6$ when $r = 2$. (a) Find a formula for F in terms of r . (b) Find r when $F = 3072$.

STEP 1 $F = \frac{k}{r^3}$ ◀ L

STEP 2 $6 = \frac{k}{8} \Rightarrow k = 48; \quad F = \frac{48}{r^3}$ ◀ I, N: (a)

STEP 4 $r^3 = \frac{48}{3072} = \frac{1}{64} \Rightarrow r = 0.25$ ◀ K: (b), cube root

ANSWER (a) $F = \frac{48}{r^3}$ (b) $r = 0.25$

✓ $48 \div 0.25^3 = 3072$ ✓

WATCH OUT

✗ $F = \frac{k}{r}$ then $k = 12$

✓ $F = \frac{k}{r^3}, k = 48$

"the cube of r " means r^3 in the Layout.

EXAM TIP

Non-calculator paper: $\frac{48}{3072} = \frac{1}{64}$ — divide top and bottom by 48.

Check yourself

calculator first • answers on page 17

3.1 Jan 2022 P1H

A is inversely proportional to C^2 .
 $A = 40$ when $C = 1.5$.
 Find C when $A = 1000$.

Answer:

3.2 Jun 2022 P2H

M varies directly as the cube of h . $M = 4$ when $h = 0.5$. Find h when $M = 500$.

Answer:

3.3 Nov 2025 P2H

T is inversely proportional to $\sqrt{m}$.
 $T = 15$ when $m = 36$.
 Find a formula.

Answer:

V4

VARIANT 4

Graphs & Tables

Read one exact pair from the graph or table, then LINK; or test $\frac{y}{x}$ and xy .

SEEN IN PAST PAPERS

1 / 17

PRACTICE QUESTIONS: 10



Your notes

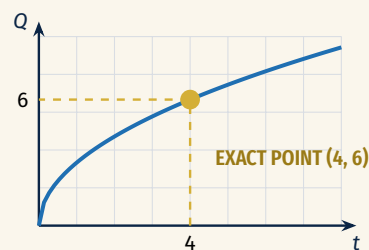
Came up in: Jun 2023 P2H Q16(a)

? What is it?**In plain words**

A graph gives you the **first pair**: pick a point that sits **exactly** on grid lines. Then LINK.

A table: if $\frac{y}{x}$ is constant it is **direct**; if xy is constant it is **inverse**.

IT LOOKS LIKE
 $Q \propto \sqrt{t}$ graph $x : 2$ 4

 8 $y : 12$ 6 3
**How do I do it?****1 Layout**From the words: $Q = k\sqrt{t}$.**2 Exact point**

A point exactly on the grid: (4, 6).

3 Insert $6 = k\sqrt{4} \Rightarrow k = 3$.**4 Check**Test a second point: $t = 1$, $Q = 3 \checkmark$.**EXACT POINT, THEN LINK** Tables: test $\frac{y}{x}$ and xy .

V4 Graphs & Tables

VARIANT 4 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"The graph shows the relationship ... find a formula"

→ Exact grid point → k

A table of x and y

→ Constant $\frac{y}{x}$: direct; constant xy : inverse

"Which graph shows $y \propto x$?"

→ Straight line through the origin

WORKED EXAMPLE PAST PAPER

June 2023 • Paper 2H • Q16(a) • 2 marks

Q is directly proportional to $\sqrt{t}$. The graph of Q against t passes through (4, 6) and (1, 3). Find a formula for Q in terms of t .

STEP 1 $Q = k\sqrt{t}$

◀ Layout

STEP 2 $t = 4, Q = 6$

◀ exact point from the graph

STEP 3 $6 = k \times 2 \Rightarrow k = 3$

◀ Insert

STEP 4 $Q = 3\sqrt{t}$

◀ New equation

ANSWER $Q = 3\sqrt{t}$

✓ $t = 1: Q = 3$ ✓

WATCH OUT

× reading (5, 6.7) off-grid

✓ use (4, 6) exactly

Estimated readings give a wrong k .

EXAM TIP

Choose a t that is a perfect square (1, 4, 9) — the root is exact.

Check yourself

calculator first • answers on page 17

4.1 Dr Eslam

x : 2, 4, 8 and y : 12, 6, 3.
Direct or inverse? Give the formula.

Answer:

4.2 Practice • Hard

$y \propto \frac{1}{x^2}$, $y = 25$ when $x = 2$. Find y for $x = 1, 5, 10$.

Answer:

4.3 Dr Eslam

A straight-line graph through the origin shows which proportion?

Answer:

V5

VARIANT 5

Two Links (Chained Proportion)

LINK each sentence, then put one formula inside the other.

SEEN IN PAST PAPERS

1 / 17

PRACTICE QUESTIONS: 4


Your notes

Came up in: Jan 2022 P2HR Q20

? What is it?

In plain words

Two $\propto$ sentences share a middle letter: $y \propto \frac{1}{\sqrt{x}}$ and $x \propto T^3$.

Substitute the second into the first:

$y = \frac{a}{\sqrt{bT^3}}$, which is just $y = \frac{c}{T^{3/2}}$ —

one new constant. Then LINK once.

IT LOOKS LIKE

$$y \propto \frac{1}{\sqrt{x}}$$

$$x \propto T^3$$

$$y = \frac{a}{\sqrt{x}}, \quad x = bT^3$$

$$y = \frac{a}{\sqrt{bT^{3/2}}}$$

$$y = \frac{c}{T^{3/2}}$$

ONE CONSTANT IS ENOUGH

How do I do it?

1 Link 1

First sentence as an equation.

2 Link 2

Second sentence as an equation.

3 Combine

Substitute; merge the constants.

4 LINK

Insert the pair, then solve.

LINK INSIDE LINK Nothing new — LINK twice.

V5 Two Links (Chained Proportion)

VARIANT 5 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DOTwo $\propto$ sentences, three letters→ **Substitute the middle letter away**"Find the exact value of T "→ **Leave as a fraction: $T = \frac{100}{9}$** $T^{3/2} = \dots$ → **Square, then cube root (or power $\frac{2}{3}$)****WORKED EXAMPLE****PAST PAPER**

January 2022 · Paper 2HR · Q20 · 4 marks

y is inversely proportional to $\sqrt{x}$; x is directly proportional to T^3 . $y = 8$ when $T = 25$. Find the exact value of T when $y = 27$.

STEP 1 $y = \frac{c}{T^{3/2}}$

◀ combine: $\sqrt{T^3} = T^{3/2}$

STEP 2 $8 = \frac{c}{125} \Rightarrow c = 1000$

◀ $25^{3/2} = 125$

STEP 3 $27 = \frac{1000}{T^{3/2}} \Rightarrow T^{3/2} = \frac{1000}{27}$

◀ substitute

STEP 4 $T = \left(\frac{10}{3}\right)^2 = \frac{100}{9}$

◀ cube root, then square

ANSWER $T = \frac{100}{9}$

$$\sqrt{\left(\frac{100}{9}\right)^{3/2}} = \frac{1000}{27} \quad \checkmark$$

**WATCH OUT**× two constants a, b to find✓ merge into one c

Only one pair is given, so only one constant can be found.

**EXAM TIP**

$T^{3/2}$: cube root first ($\frac{10}{3}$), then square — the numbers stay small.

**Check yourself**

calculator first · answers on page 17

5.1

Dr Eslam

$y \propto z$, $y = 12$ when $z = 4$.
 $z \propto \frac{1}{x}$, $z = 3$ when $x = 2$.
 Find y in terms of x .

Answer:

5.2

Practice · Very hard

$h = \frac{60}{p}$ and $p = 0.5\sqrt{t}$.
 Find h in terms of t .

Answer:

5.3

Dr Eslam

$a \propto b^2$, $a = 8$ when
 $b = 2$. $b \propto \frac{1}{c}$, $b = 3$
 when $c = 4$. Find a in
 terms of c .

Answer:

V6

VARIANT 6

Percentage Change

Turn the change into a multiplier and push it through the power.

SEEN IN PAST PAPERS

1 / 17

PRACTICE QUESTIONS: 5



Your notes

Came up in: Jun 2023 P2H Q16(b)

? What is it?**In plain words**

A 20% increase is $\times 1.2$. If $Q = 3\sqrt{t}$, then $\sqrt{t}$ must go up $\times 1.2$, so t goes up $\times 1.2^2 = 1.44$: a **44% increase**.

k never matters — only the **power**.

IT LOOKS LIKE $Q \uparrow 20\%$ $x \downarrow 10\%$ with $y \propto x^3$

$$Q \propto \sqrt{t} \iff t \propto Q^2$$

$$Q \times 1.2 \Rightarrow t \times 1.2^2$$

$$t \times 1.44: +44\%$$

MULTIPLIER THROUGH THE POWER**How do I do it?****1 Multiplier**+20% $\rightarrow$ 1.2, -10% $\rightarrow$ 0.9.**2 Power**

Raise it to the power linking the two letters.

3 Inverse?Inverse: $1 \div$ the result.**4 Back to %**1.44 $\rightarrow$ +44%; 0.729 $\rightarrow$ -27.1%.**MULTIPLIER THROUGH THE POWER** k cancels — ignore it.

V6 Percentage Change

VARIANT 6 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Q is increased by 20%. Find the percentage increase in t"

$$\rightarrow t \propto Q^2: 1.2^2$$

"x is decreased by 10%", $y \propto x^3$

$$\rightarrow 0.9^3 = 0.729$$

Inversely proportional

$$\rightarrow \frac{1}{\text{multiplier}}$$

WORKED EXAMPLE **PAST PAPER**

June 2023 · Paper 2H · Q16(b) · 2 marks

$Q = 3\sqrt{t}$. Q is increased by 20%. Find the percentage increase in t.

STEP 1 $t = \left(\frac{Q}{3}\right)^2 \Rightarrow t \propto Q^2$

◀ the power from t to Q

STEP 2 $Q \times 1.2$

◀ multiplier

STEP 3 $t \times 1.2^2 = t \times 1.44$

◀ through the power

STEP 4 $1.44 \rightarrow +44\%$

◀ back to a percentage

ANSWER 44%

$$\checkmark Q = 6 \rightarrow 7.2: t = 4 \rightarrow 5.76, \times 1.44 \checkmark$$

WATCH OUT

$$\times 20\% \times 2 = 40\%$$

$$\checkmark 1.2^2 = 1.44: 44\%$$

Percentages multiply, they don't add.

EXAM TIP

Not sure? Pick numbers: $t = 4$, find Q, raise Q by 20%, find the new t.

Check yourself

calculator first · answers on page 17

6.1 Dr Eslam

$y \propto x^3$. x is decreased by 10%. Find the percentage decrease in y.

Answer:

6.2 Practice · Very hard

$y \propto x^2$. x is increased by 15%. Find the percentage increase in y.

Answer:

6.3 Practice · Very hard

P is inversely proportional to V. V is increased by 25%. Percentage change in P?

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 £5.44

1.2 24 minutes

1.3 250 km

2.1 $y = 35$

2.2 $w = 4t$

2.3 $d = 20$

3.1 $C = 0.3$

3.2 $h = 2.5$

3.3 $T = \frac{90}{\sqrt{m}}$

4.1 $y = \frac{24}{x}$

4.2 100, 4, 1

4.3 $y \propto x$ (direct)

5.1 $y = \frac{18}{x}$

5.2 $h = \frac{120}{\sqrt{t}}$

5.3 $a = \frac{288}{c^2}$

6.1 27.1% decrease

6.2 32.25%

6.3 20% decrease

THE WHOLE TOPIC ON ONE PAGE

Proportion cheat sheet



Your notes
V1 In words

$$6 \times 10 \div 4 = 15$$

Inverse: total job first.

V2 Direct & inverse

$$y = kx, \quad y = \frac{k}{x}$$

L-I-N-K.

V3 Powers, roots 14/17

$$F = \frac{48}{r^3}$$

Power on the letter; undo to find it.

V4 Graphs, tables 1/17

$$(4, 6) \rightarrow k = 3$$

Exact point; test $\frac{y}{x}$, xy .**V5 Two links 1/17**

$$y = \frac{c}{T^{3/2}}$$

Substitute; one constant.

V6 % change 1/17

$$1.2^2 = 1.44$$

Multiplier through the power.


TOP 3 LOST MARKS 1. forgetting the power in the Layout $\times$ 2. rounding $k \times$ 3. adding percentages $\times$
BOTTOM LINE Layout • Insert • New equation • Keep solving • power on the letter • one constant.